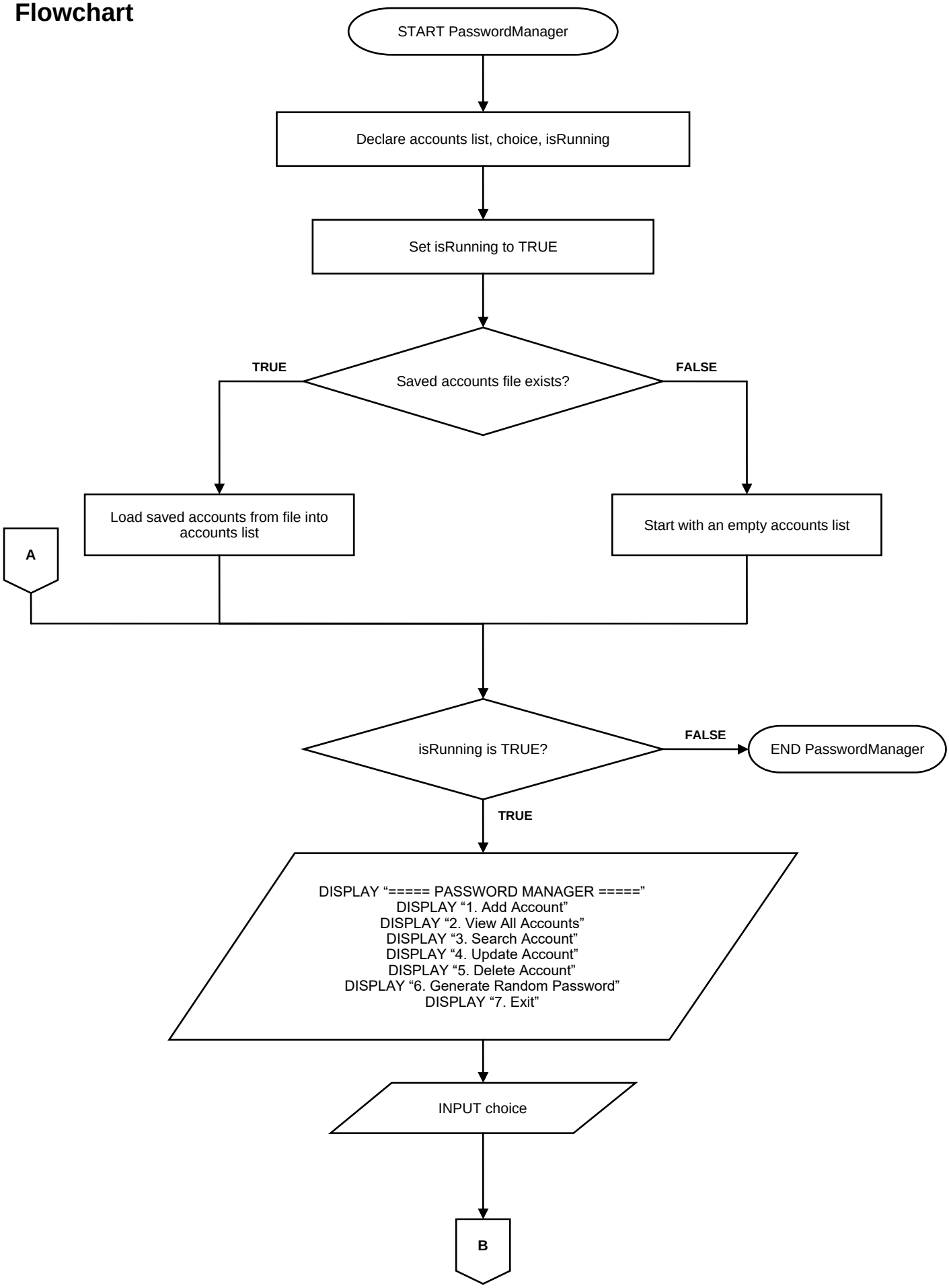
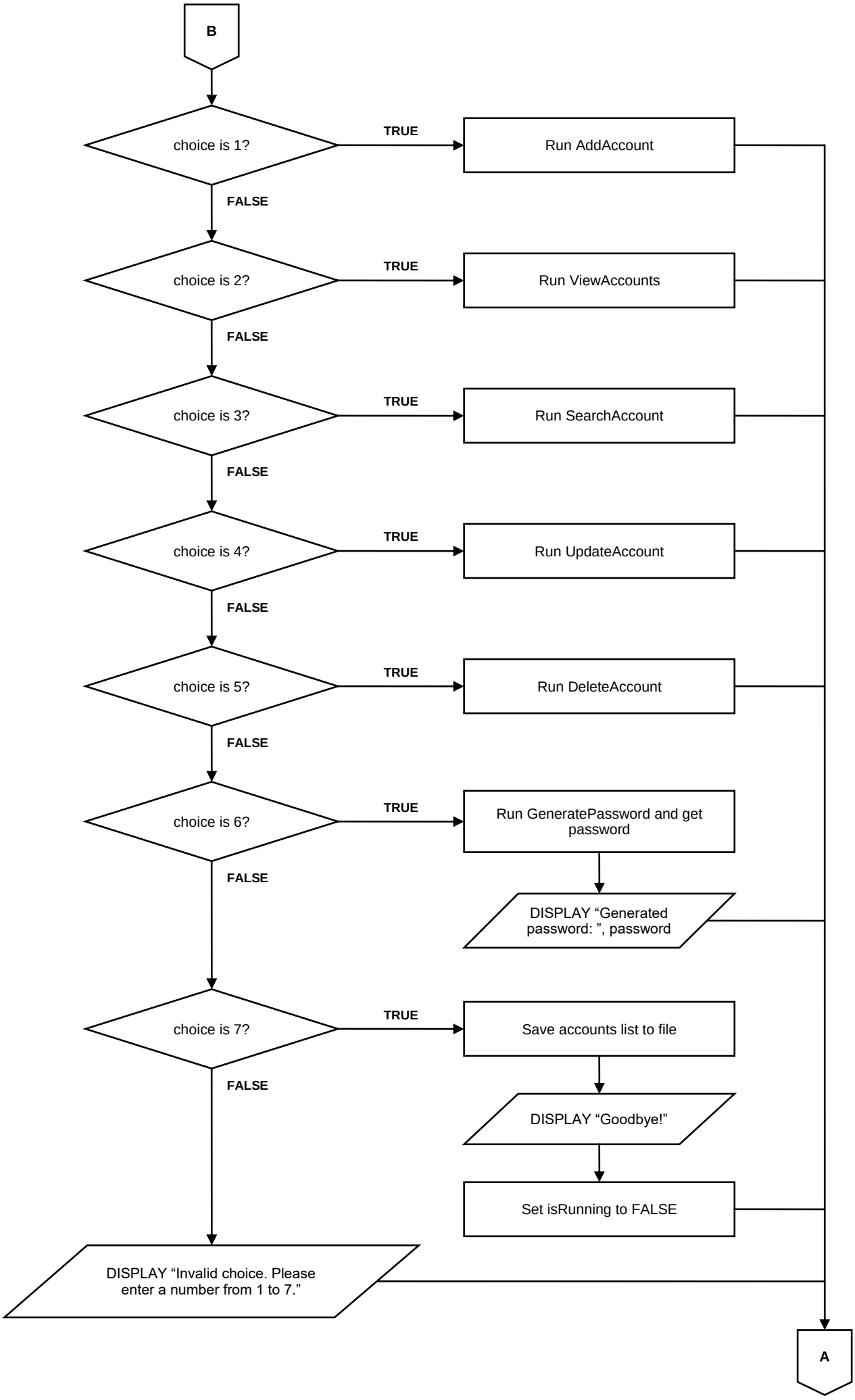
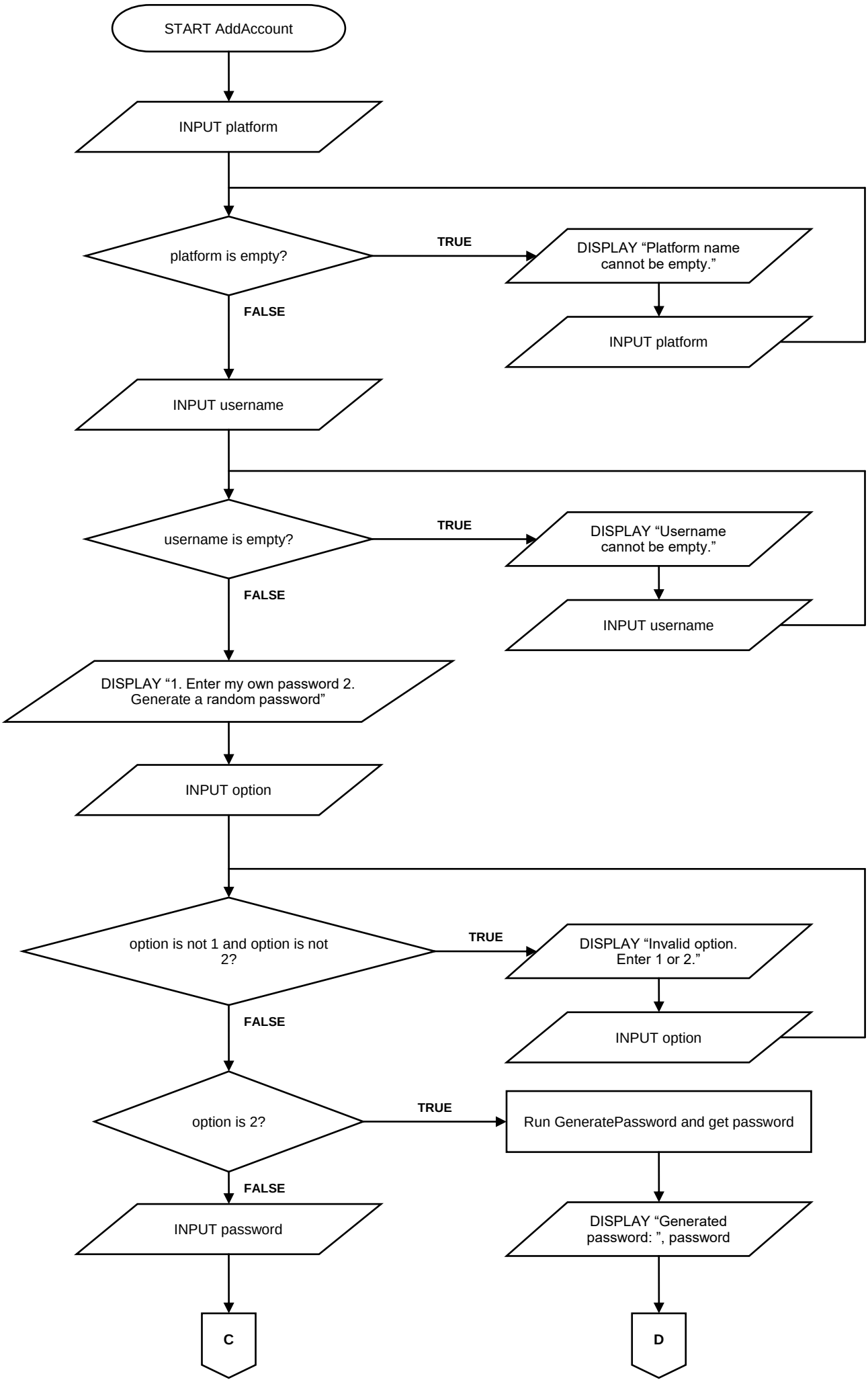
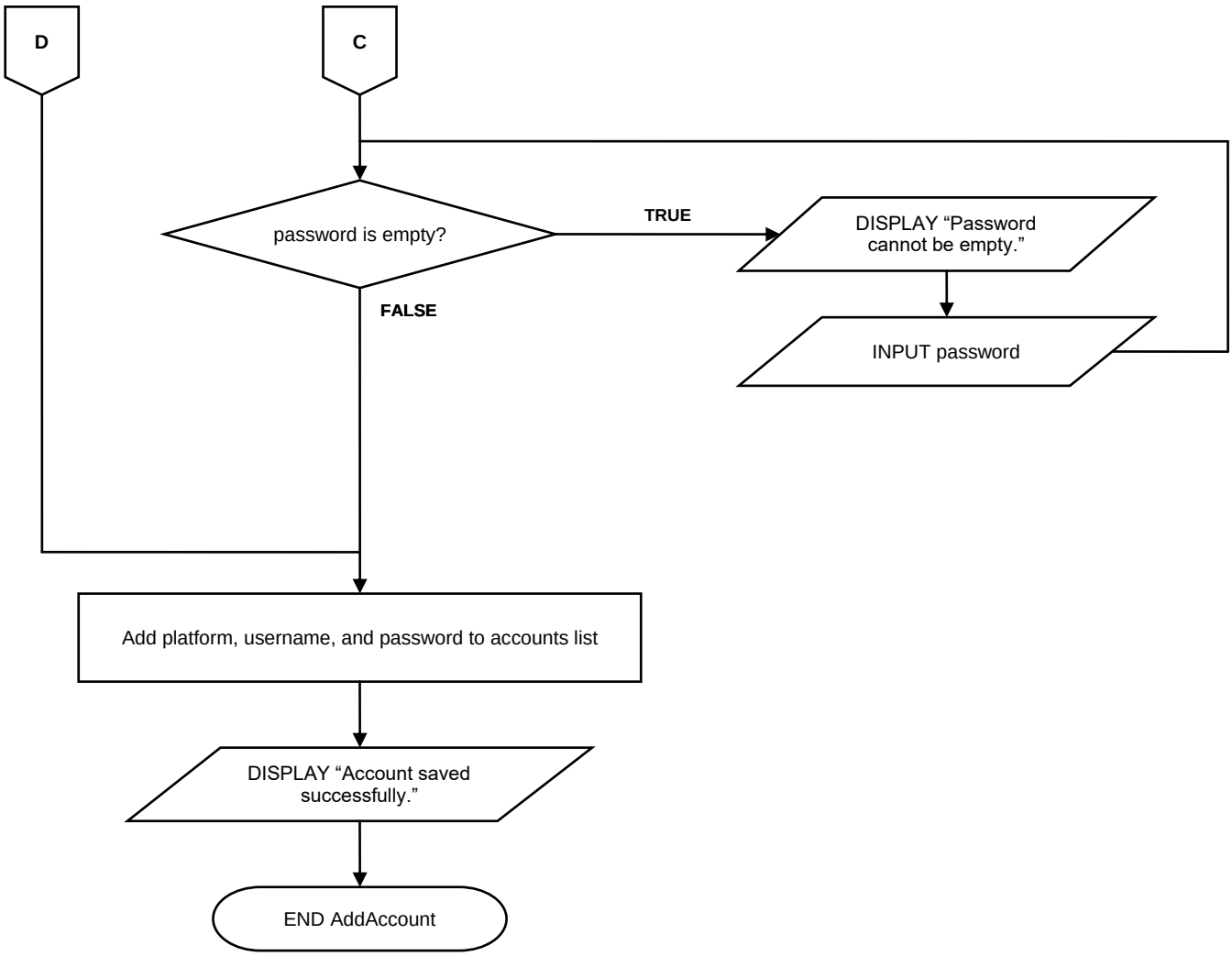
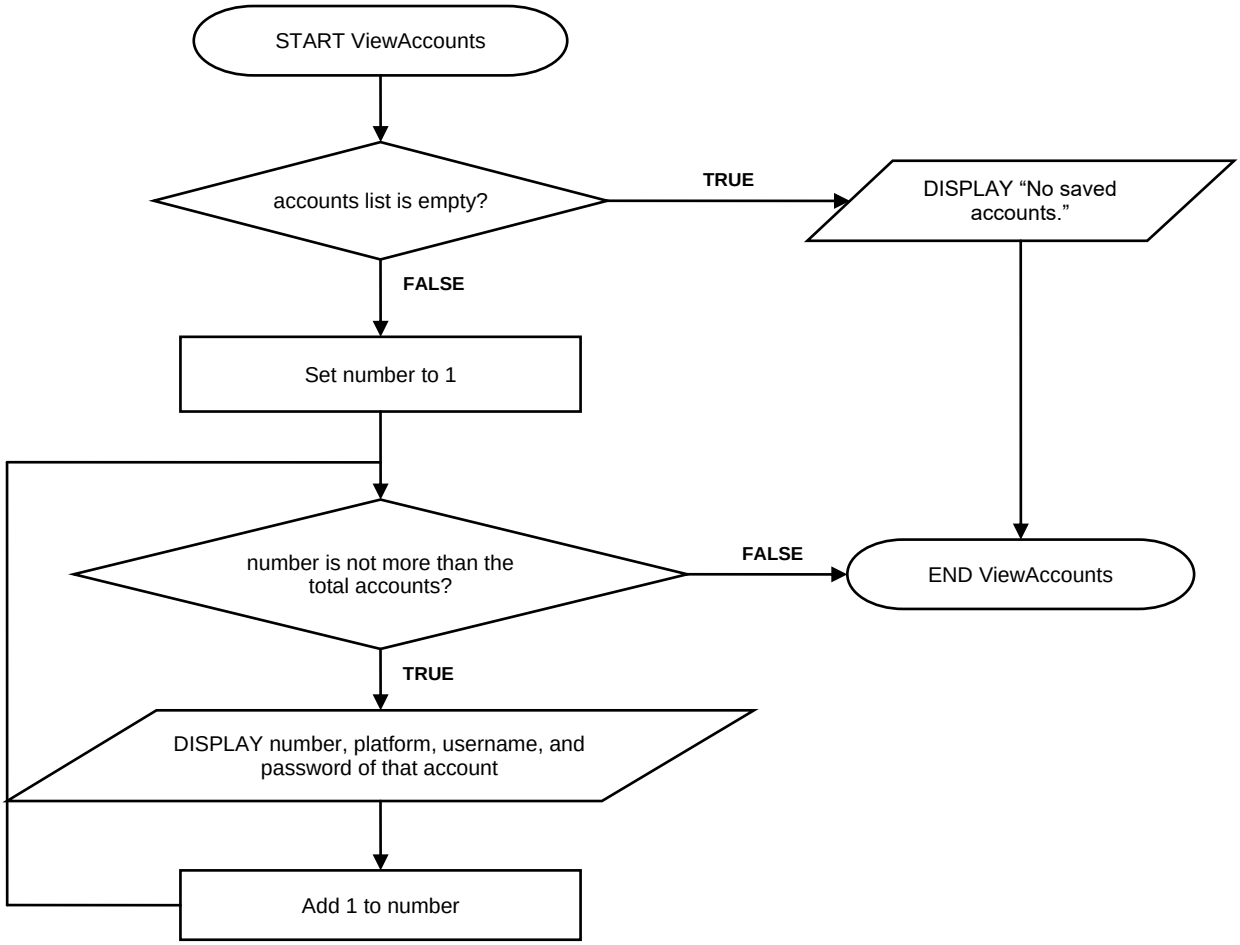
Flowchart

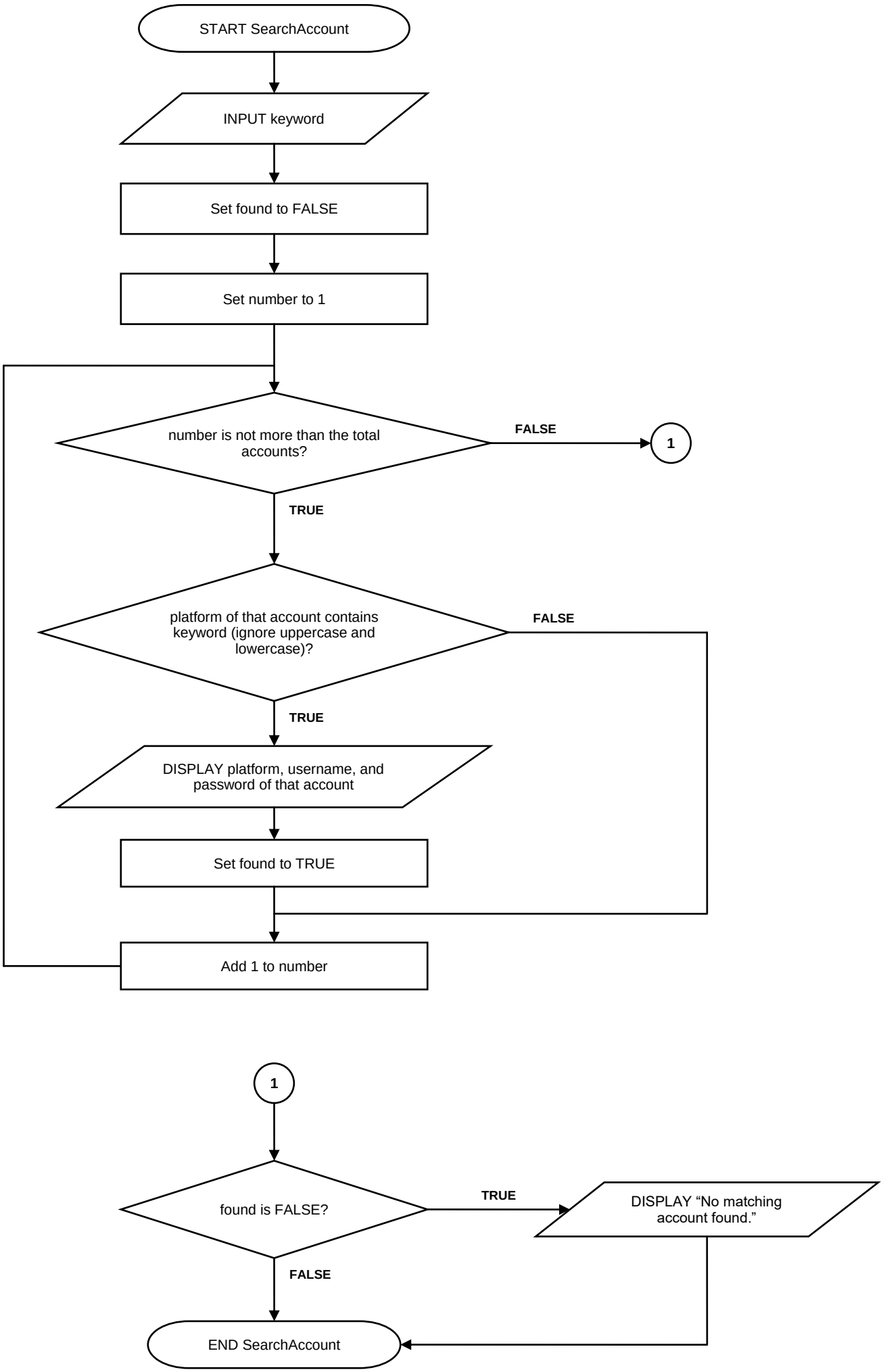

START PasswordManager
Declare accounts list, choice, isRunning
Set isRunning to TRUE
Saved accounts file exists?
TRUE
FALSE
Load saved accounts from file into accounts list
Start with an empty accounts list
A
isRunning is TRUE?
FALSE
END PasswordManager
TRUE
DISPLAY "===== PASSWORD MANAGER ====="
DISPLAY "1. Add Account"
DISPLAY "2. View All Accounts"
DISPLAY "3. Search Account"
DISPLAY "4. Update Account"
DISPLAY "5. Delete Account"
DISPLAY "6. Generate Random Password"
DISPLAY "7. Exit"
INPUT choice
B

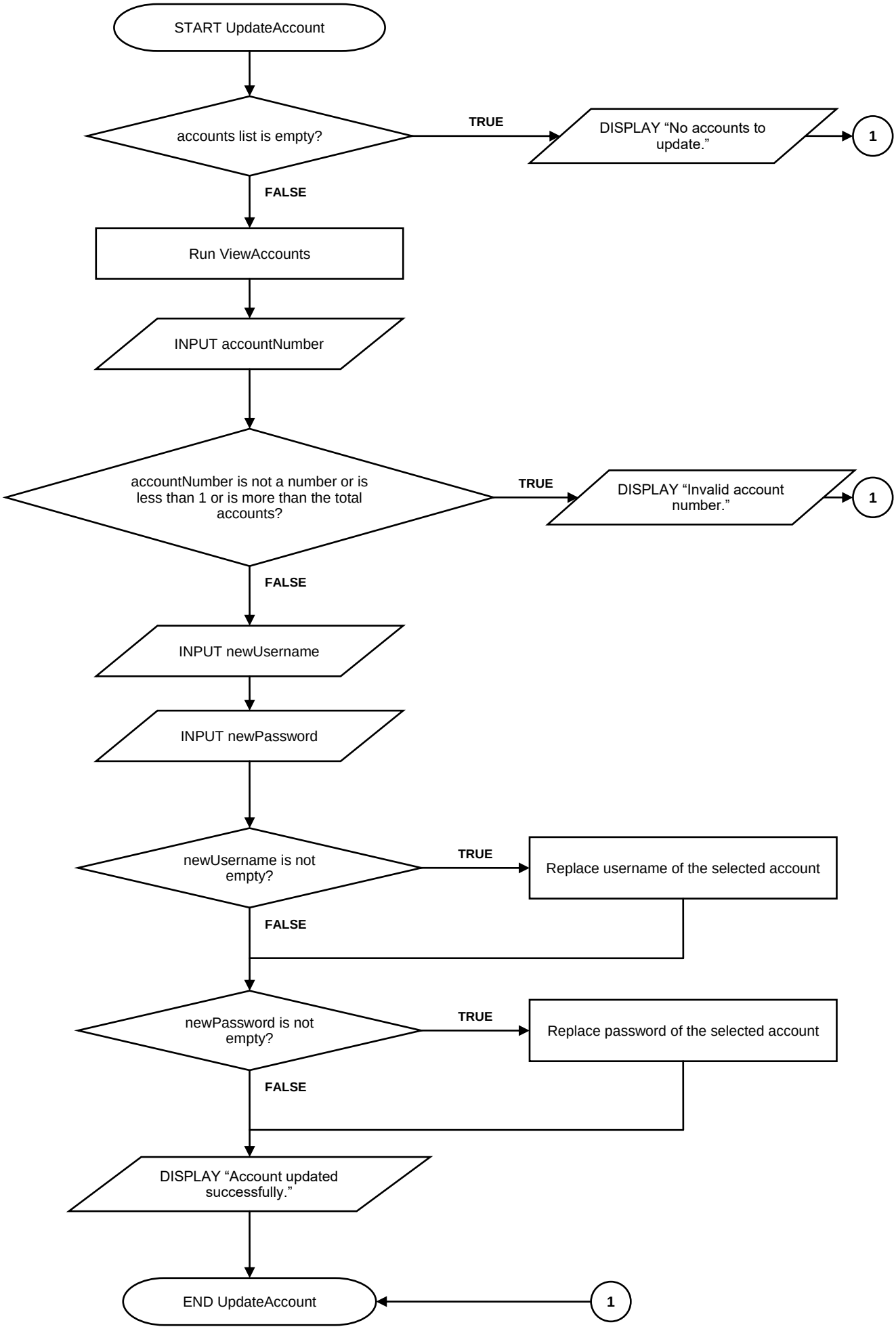

B
choice is 1?
TRUE
Run AddAccount
FALSE
choice is 2?
TRUE
Run ViewAccounts
FALSE
choice is 3?
TRUE
Run SearchAccount
FALSE
choice is 4?
TRUE
Run UpdateAccount
FALSE
choice is 5?
TRUE
Run DeleteAccount
FALSE
choice is 6?
TRUE
Run GeneratePassword and get password
DISPLAY “Generated password: ”, password
FALSE
choice is 7?
TRUE
Save accounts list to file
DISPLAY “Goodbye!”
Set isRunning to FALSE
FALSE
DISPLAY “Invalid choice. Please enter a number from 1 to 7.”
A

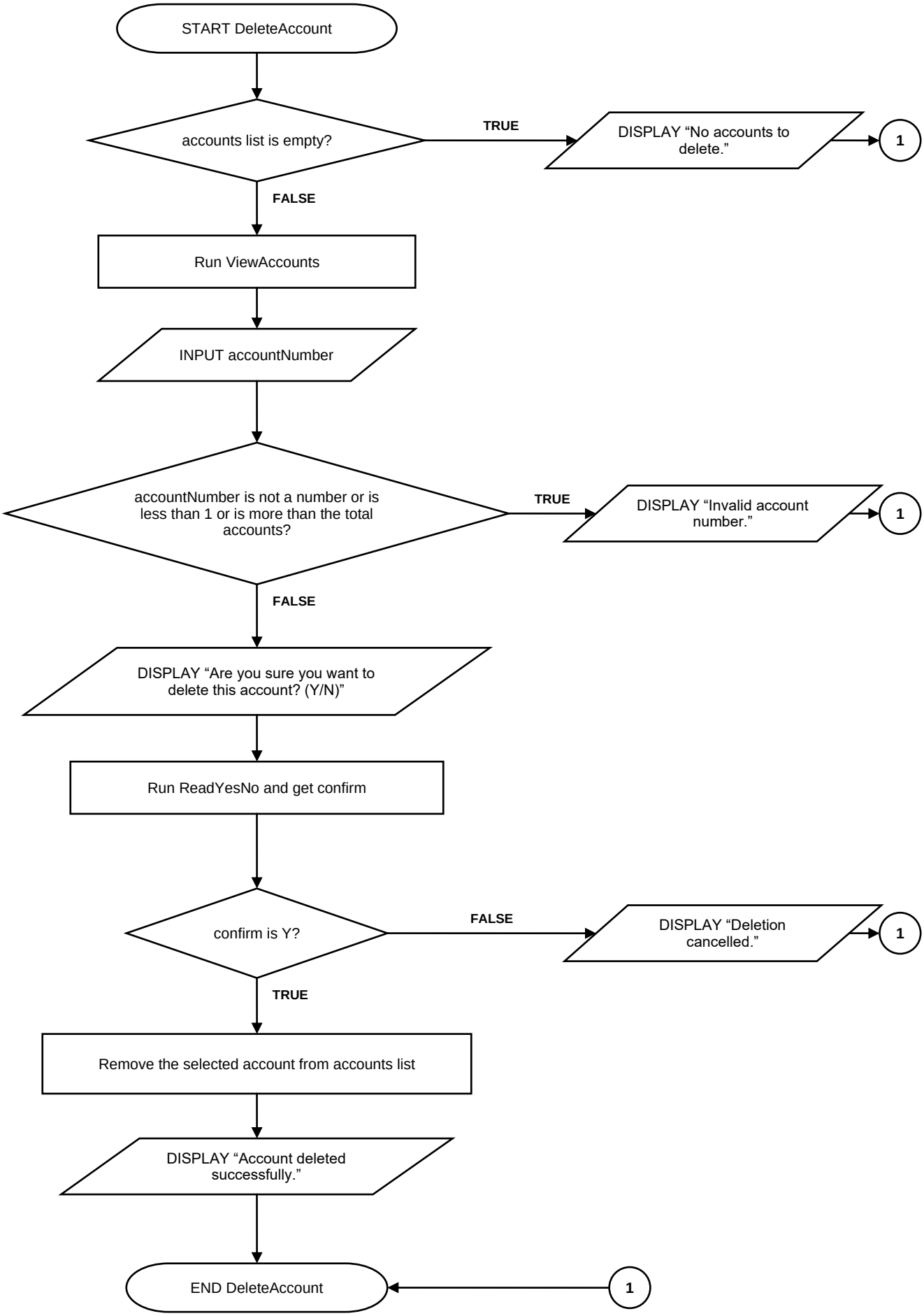

START AddAccount
INPUT platform
platform is empty?
TRUE
DISPLAY “Platform name cannot be empty.”
INPUT platform
FALSE
INPUT username
username is empty?
TRUE
DISPLAY “Username cannot be empty.”
INPUT username
FALSE
DISPLAY “1. Enter my own password 2. Generate a random password”
INPUT option
option is not 1 and option is not 2?
TRUE
DISPLAY “Invalid option. Enter 1 or 2.”
INPUT option
FALSE
option is 2?
TRUE
Run GeneratePassword and get password
FALSE
INPUT password
DISPLAY “Generated password: ”, password
C
D

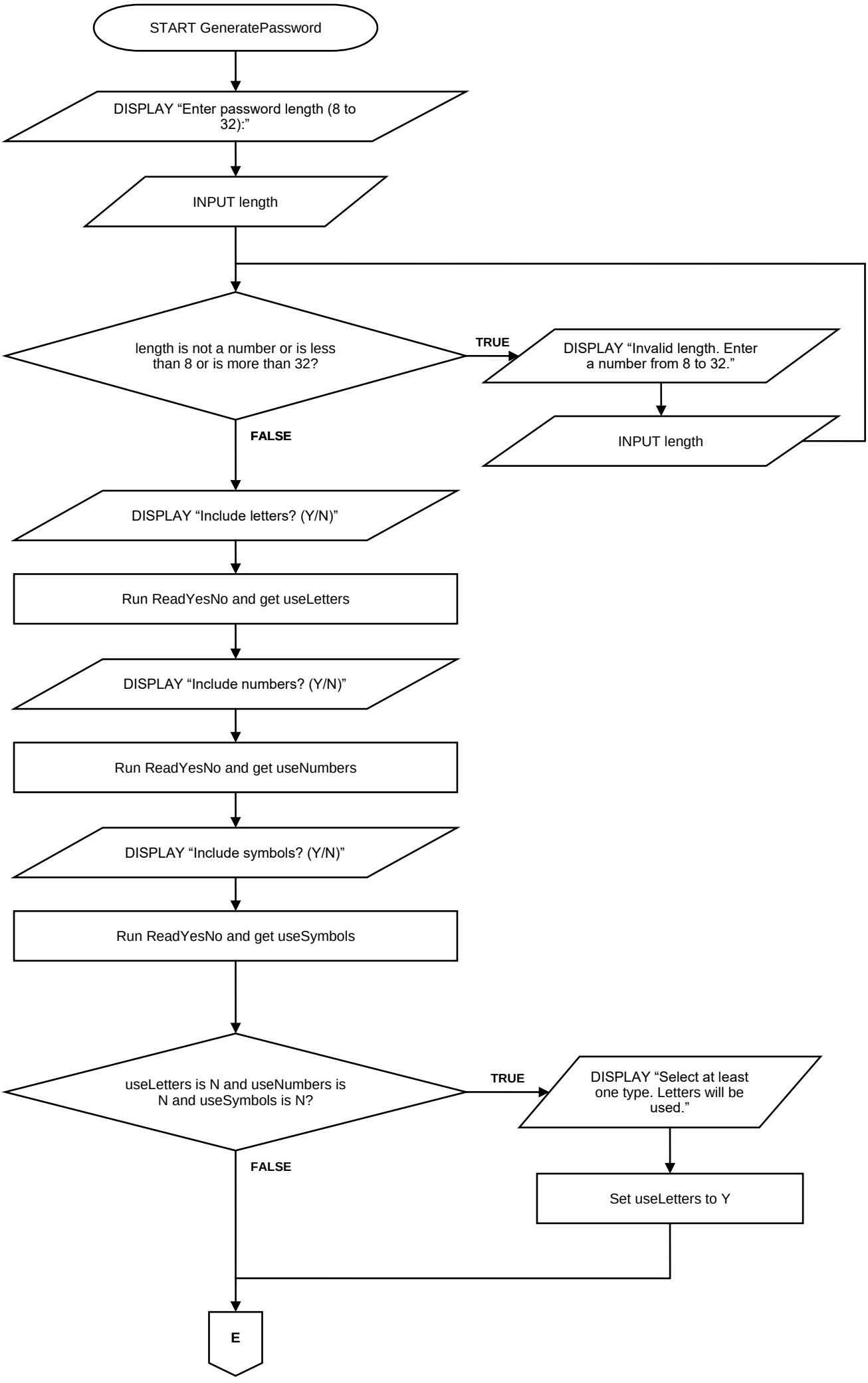

D
C
password is empty?
TRUE
DISPLAY “Password cannot be empty.”
INPUT password
FALSE
Add platform, username, and password to accounts list
DISPLAY “Account saved successfully.”
END AddAccount

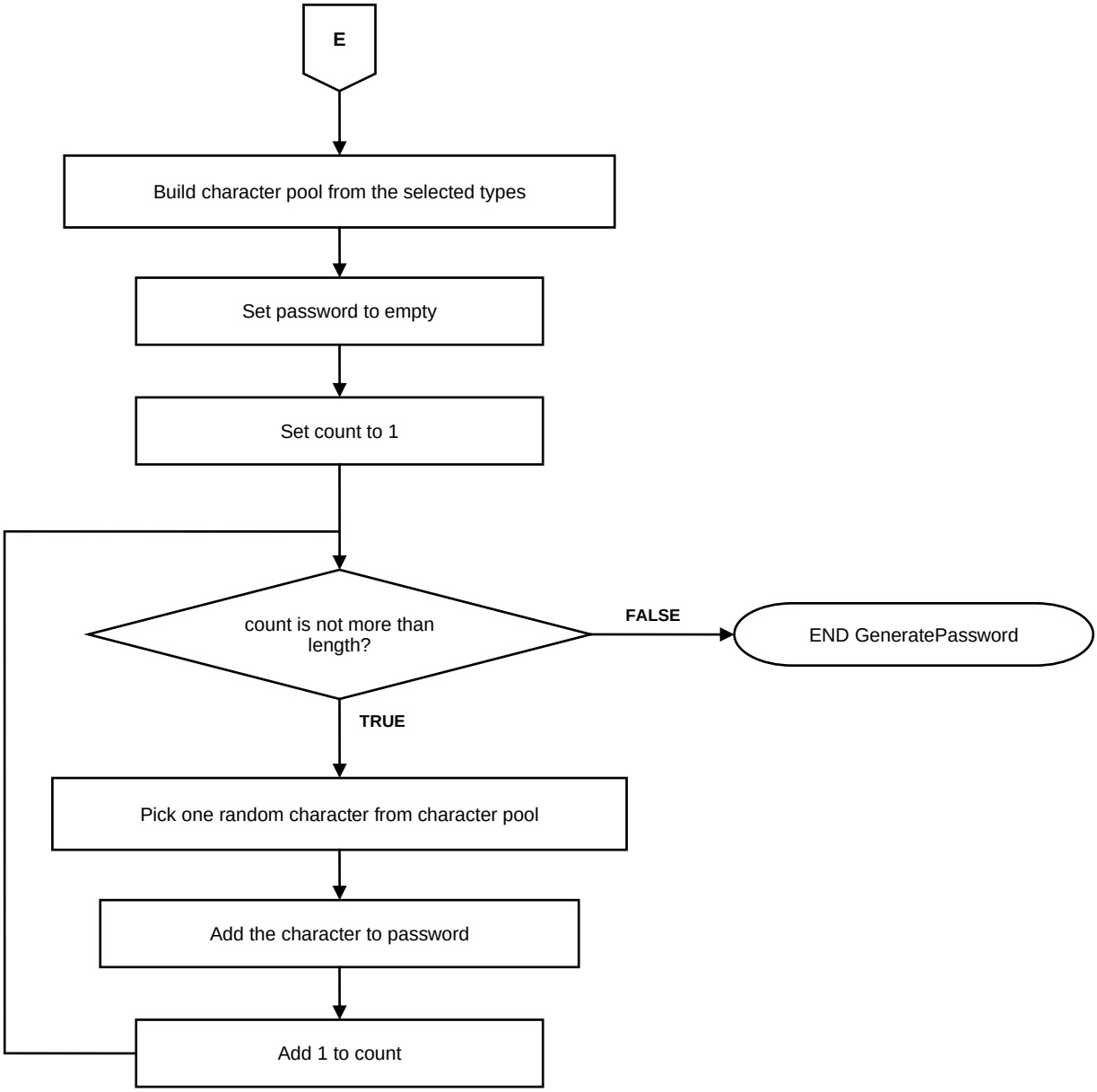

START ViewAccounts
accounts list is empty?
TRUE
DISPLAY “No saved accounts.”
FALSE
Set number to 1
number is not more than the total accounts?
FALSE
END ViewAccounts
TRUE
DISPLAY number, platform, username, and password of that account
Add 1 to number

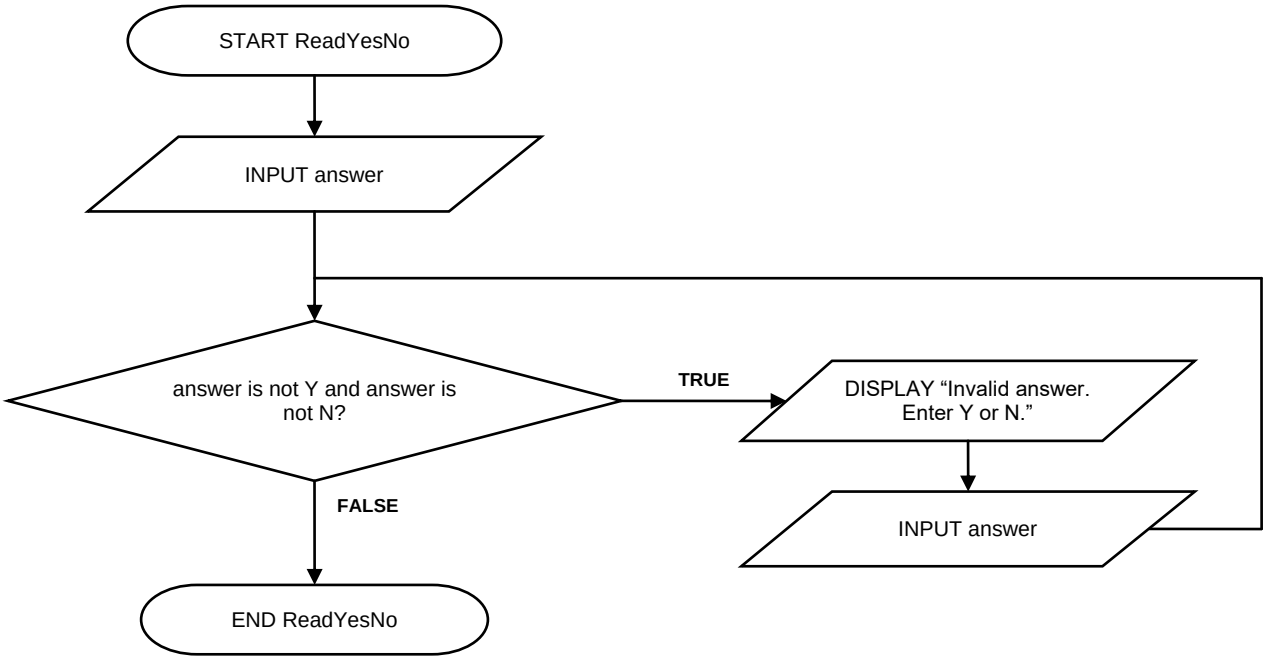

START SearchAccount
INPUT keyword
Set found to FALSE
Set number to 1
number is not more than the total accounts?
FALSE
1
TRUE
platform of that account contains keyword (ignore uppercase and lowercase)?
FALSE
TRUE
DISPLAY platform, username, and password of that account
Set found to TRUE
Add 1 to number
1
found is FALSE?
TRUE
DISPLAY “No matching account found.”
FALSE
END SearchAccount


START UpdateAccount
accounts list is empty?
TRUE
DISPLAY “No accounts to update.”
1
FALSE
Run ViewAccounts
INPUT accountNumber
accountNumber is not a number or is less than 1 or is more than the total accounts?
TRUE
DISPLAY “Invalid account number.”
1
FALSE
INPUT newUsername
INPUT newPassword
newUsername is not empty?
TRUE
Replace username of the selected account
FALSE
newPassword is not empty?
TRUE
Replace password of the selected account
FALSE
DISPLAY “Account updated successfully.”
END UpdateAccount
1


START DeleteAccount
accounts list is empty?
TRUE
DISPLAY “No accounts to delete.”
1
FALSE
Run ViewAccounts
INPUT accountNumber
accountNumber is not a number or is less than 1 or is more than the total accounts?
TRUE
DISPLAY “Invalid account number.”
1
FALSE
DISPLAY “Are you sure you want to delete this account? (Y/N)”
Run ReadYesNo and get confirm
confirm is Y?
FALSE
DISPLAY “Deletion cancelled.”
1
TRUE
Remove the selected account from accounts list
DISPLAY “Account deleted successfully.”
END DeleteAccount
1


START GeneratePassword
DISPLAY “Enter password length (8 to 32):”
INPUT length
length is not a number or is less than 8 or is more than 32?
TRUE
DISPLAY “Invalid length. Enter a number from 8 to 32.”
INPUT length
FALSE
DISPLAY “Include letters? (Y/N)”
Run ReadYesNo and get useLetters
DISPLAY “Include numbers? (Y/N)”
Run ReadYesNo and get useNumbers
DISPLAY “Include symbols? (Y/N)”
Run ReadYesNo and get useSymbols
useLetters is N and useNumbers is N and useSymbols is N?
TRUE
DISPLAY “Select at least one type. Letters will be used.”
Set useLetters to Y
FALSE
E


E
Build character pool from the selected types
Set password to empty
Set count to 1
count is not more than length?
FALSE
END GeneratePassword
TRUE
Pick one random character from character pool
Add the character to password
Add 1 to count


START ReadYesNo
INPUT answer
answer is not Y and answer is not N?
TRUE
DISPLAY “Invalid answer. Enter Y or N.”
INPUT answer
FALSE
END ReadYesNo